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WO-100-03 — Replace spool SP02 of 6"-P150-P-1001

Area	A100 Tank farm	**Task**	SR Spool replacement
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**T-101** — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	6"-P150-P-1001
Isometrics	ISO-6"-P150-P-1001 (generated); owner's drawings ISO-P-1001		
Permits	General work permit, Hot work permit		

Reason

External corrosion found on the tank outlet line at the first bend.

Scope of work

1. Isolate per the isolation list; drain, vent and purge 6"-P150-P-1001 between the isolations; confirm gas-free for hot work.
2. Remove spool 6"-P150-P-1001-SP02 (7 items, 37'-0" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-6"-P150-P-1001 with the same items (parts list), welding to WS-01; 6 shop welds.
4. Install the spool; make 2 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 210 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 6"-P150-P-1001/el7, 6"-P150-P-1001/p8, 6"-P150-P-1001/el9, 6"-P150-P-1001/p10, 6"-P150-P-1001/el11, 6"-P150-P-1001/p12, 6"-P150-P-1001/el13

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1001	6"-P150-P-1001	6"-P150-P-1001	5	1 error(s), 4 warning(s) — see below

ISO-P-1001 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1001	5	error	sheet 5: CONT'D ON ISO-P-1002 prints E 154'-5" N -12'-6" EL 102'-0" for E 154'-5" N -11'-6" EL 102'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 6"-P150-P-1001 meets 6"-P150-P-1002 (E 154'-5" N -11'-6" EL 102'-0")
ISO-P-1001	5	warning	W21 (6"-P150-P-1001/W21, sheet 5) is labelled again on sheet 4; it is 58.0 in from the nearest break
ISO-P-1001	4	warning	branch weld 3/4"-P300-I-7030/W1 (E 151'-6" N 0" EL 117'-5", sheet 4) has no label
ISO-P-1001	4	warning	branch weld 3/4"-P300-I-7001/W1 (E 152'-6" N 0" EL 117'-5", sheet 4) has no label
ISO-P-1001	5	warning	TO label 'P-101A NOZ. S' is not on the drawing

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
6"-P150-P-1001/el7	item	ELBOW 90° R = 5/3 OD 6" P150	6"-P150-P-1001	E 33'-7" N 26'-6" EL 101'-11"
6"-P150-P-1001/p8	item	PIPE 6" P150 BEVELLED ENDS	6"-P150-P-1001	E 34'-0" N 26'-6" EL 107'-6"
6"-P150-P-1001/el9	item	ELBOW 90° R = 5/3 OD 6" P150	6"-P150-P-1001	E 34'-0" N 26'-1" EL 113'-1"
6"-P150-P-1001/p10	item	PIPE 6" P150 BEVELLED ENDS	6"-P150-P-1001	E 34'-0" N 13'-3" EL 113'-6"
6"-P150-P-1001/el11	item	ELBOW 90° R = 5/3 OD 6" P150	6"-P150-P-1001	E 34'-0" N 5" EL 113'-11"
6"-P150-P-1001/p12	item	PIPE 6" P150 BEVELLED ENDS	6"-P150-P-1001	E 34'-0" N 0.000" EL 115'-3"

Tag	Kind	Description	Line	Location
6"-P150-P-1001/el13	item	ELBOW 90° R = 5/3 OD 6" P150	6"-P150-P-1001	E 34'-5" N 0.000" EL 116'-7"
6"-P150-P-1001/A2	item	VALVE, GATE, WEDGE, OS&Y 6" P150	6"-P150-P-1001	E 32'-2-15/16" N 26'-6" EL 101'-6"
6"-P150-P-1001/A5	item	FLANGE WELD-NECK 6" P150 RAISED FACE	6"-P150-P-1001	E 33'-1/16" N 26'-6" EL 101'-6"
6"-P150-P-1001/B7	item	VALVE, GATE, WEDGE, OS&Y 6" P150	6"-P150-P-1001	E 154'-0" N -13'-5-1/8" EL 102'-0"
6"-P150-P-1001/B10	item	FLANGE WELD-NECK 6" P150 RAISED FACE	6"-P150-P-1001	E 154'-0" N -12'-8-1/16" EL 102'-0"
6"-P150-P-1001/W1	weld	Butt weld 6" P150 (field)	6"-P150-P-1001	E 33'-2" N 26'-6" EL 101'-6"
6"-P150-P-1001/W2	weld	Butt weld 6" P150 (shop)	6"-P150-P-1001	E 34'-0" N 26'-6" EL 102'-4"
6"-P150-P-1001/W3	weld	Butt weld 6" P150 (shop)	6"-P150-P-1001	E 34'-0" N 26'-6" EL 112'-8"
6"-P150-P-1001/W4	weld	Butt weld 6" P150 (shop)	6"-P150-P-1001	E 34'-0" N 25'-8" EL 113'-6"
6"-P150-P-1001/W5	weld	Butt weld 6" P150 (shop)	6"-P150-P-1001	E 34'-0" N 10" EL 113'-6"
6"-P150-P-1001/W6	weld	Butt weld 6" P150 (shop)	6"-P150-P-1001	E 34'-0" N 0.000" EL 114'-4"
6"-P150-P-1001/W7	weld	Butt weld 6" P150 (shop)	6"-P150-P-1001	E 34'-0" N 0.000" EL 116'-2"
6"-P150-P-1001/W8	weld	Butt weld 6" P150 (field)	6"-P150-P-1001	E 34'-10" N 0.000" EL 117'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	6"-P150-P-1001/A2	6"-P150-P-1001	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/A2)	E 32'-2-15/16" N 26'-6" EL 101'-6"		
2	blind	6"-P150-P-1001/A5	6"-P150-P-1001	Fit a line blind between 6"-P150-P-1001/A2 and FLANGE WELD-NECK 6" P150 RAISED FACE (6"-P150-P-1001/A5), work side	E 32'-10-1/8" N 26'-6" EL 101'-6"		
3	valve	6"-P150-P-1001/B7	6"-P150-P-1001	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/B7)	E 154'-0" N -13'-5-1/8" EL 102'-0"		
4	blind	6"-P150-P-1001/B10	6"-P150-P-1001	Fit a line blind between 6"-P150-P-1001/B7 and FLANGE WELD-NECK 6" P150 RAISED FACE (6"-P150-P-1001/B10), work side	E 154'-0" N -12'-10" EL 102'-0"		

Parts

Description	Qty	Replaces
ELBOW 90° R = 5/3 OD 6" P150	4	
PIPE 6" P150 BEVELLED ENDS	3	

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-100-03-W01	6"-P150-P-1001/W1	ISO-P-1001 sh 1 W1	6"-P150-P-1001	6" P150	0.21875 in	field	WS-01	VT + RT	E 33'-2" N 26'-6" EL 101'-6"		
WO-100-03-W02	6"-P150-P-1001/W2	ISO-P-1001 sh 1 W2	6"-P150-P-1001	6" P150	0.21875 in	shop	WS-01	VT + RT	E 34'-0" N 26'-6" EL 102'-4"		
WO-100-03-W03	6"-P150-P-1001/W3	ISO-P-1001 sh 1 W3	6"-P150-P-1001	6" P150	0.21875 in	shop	WS-01	VT + RT	E 34'-0" N 26'-6" EL 112'-8"		
WO-100-03-W04	6"-P150-P-1001/W4	ISO-P-1001 sh 1 W4	6"-P150-P-1001	6" P150	0.21875 in	shop	WS-01	VT + RT	E 34'-0" N 25'-8" EL 113'-6"		
WO-100-03-W05	6"-P150-P-1001/W5	ISO-P-1001 sh 1 W5	6"-P150-P-1001	6" P150	0.21875 in	shop	WS-01	VT + RT	E 34'-0" N 10" EL 113'-6"		
WO-100-03-W06	6"-P150-P-1001/W6	ISO-P-1001 sh 1 W6	6"-P150-P-1001	6" P150	0.21875 in	shop	WS-01	VT + RT	E 34'-0" N 0.000" EL 114'-4"		
WO-100-03-W07	6"-P150-P-1001/W7	ISO-P-1001 sh 1 W7	6"-P150-P-1001	6" P150	0.21875 in	shop	WS-01	VT + RT	E 34'-0" N 0.000" EL 116'-2"		
WO-100-03-W08	6"-P150-P-1001/W8	ISO-P-1001 sh 1 W8	6"-P150-P-1001	6" P150	0.21875 in	field	WS-01	VT + RT	E 34'-10" N 0.000" EL 117'-0"		

Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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Scope	Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in
Joint	Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4)
Process	Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc.
Filler	Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen)
Preheat	≥ 30 °F above dew point and ≥ 60 °F, 3 in each side

Interpass	≤ 450 °F
Post-weld heat treatment	Not specified (owner to state)
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4

NDE

Method	Extent	Welds	Basis
VT	100 % of new welds, before and after welding	WO-100-03-W01, WO-100-03-W02, WO-100-03-W03, WO-100-03-W04, WO-100-03-W05, WO-100-03-W06, WO-100-03-W07, WO-100-03-W08	SW-NDE-1
RT (UT where RT is impractical)	100 % of new welds (class H150, high consequence)	WO-100-03-W01, WO-100-03-W02, WO-100-03-W03, WO-100-03-W04, WO-100-03-W05, WO-100-03-W06, WO-100-03-W07, WO-100-03-W08	SW-NDE-2

Test — Hydrostatic test

Description	Test boundary between the isolations; relief valves blinded or removed
Pressure	210 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

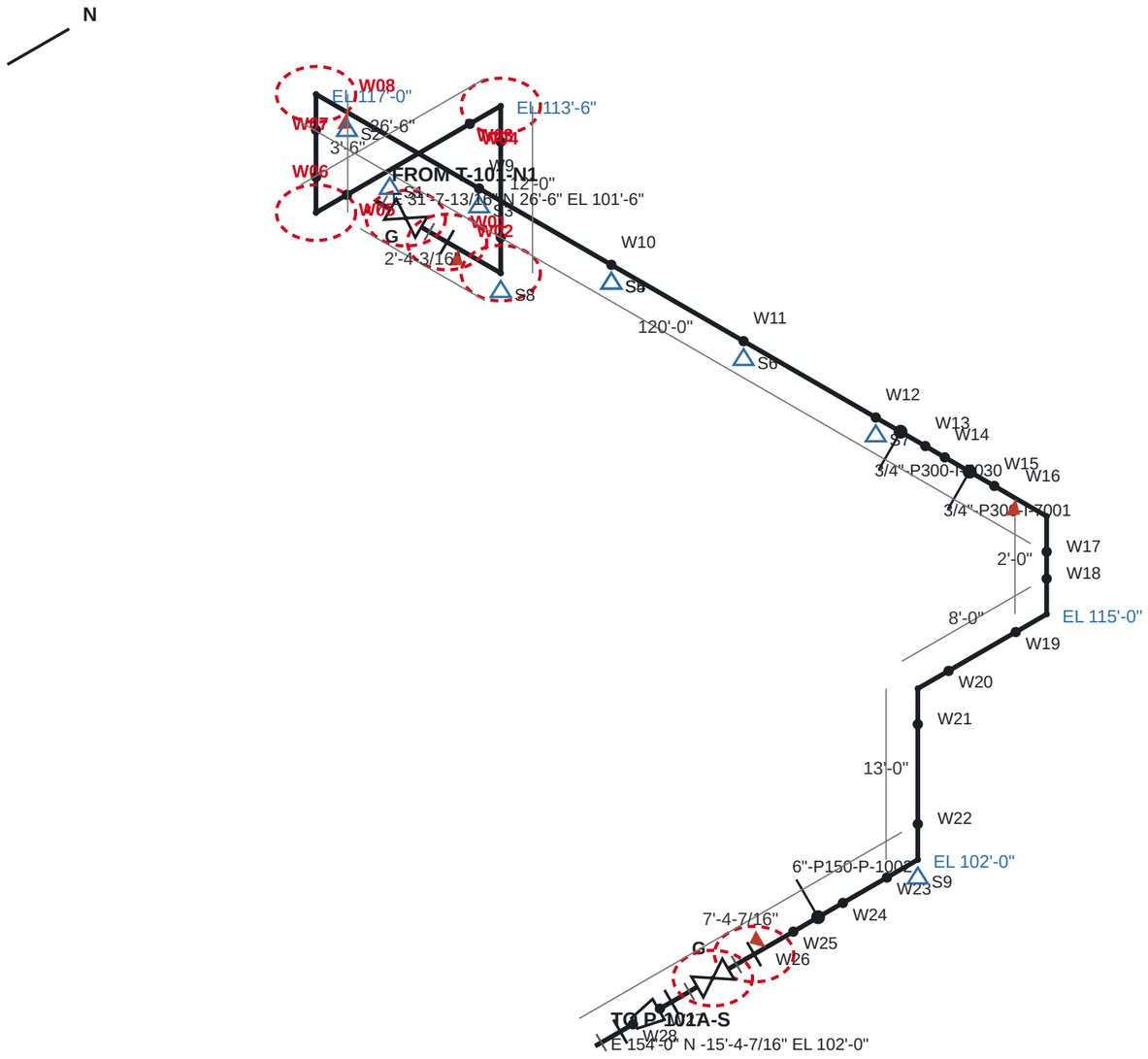
Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Line drained, vented and purged; gas-free for hot work	Owner hot-work procedure	Gas test within owner limits	Gas test record, hot work permit	H	H	—				
3	Replacement materials verified against the parts list (size, class, grade, heat marking)	Parts list, SW-LC-3	All items match the line class	Material receiving record	R	R	R				
4	Welding specification WS-01 and welder identification available at the work	WS-01, SW-WPS-3	Specification on site; every welder identified	Welder list	R	R	R				
5	Fit-up of every new weld: root opening and alignment within the root face	SW-WPS-3, SW-WD-4	Root opening 1/8 in; bores aligned within 1/16 in	Fit-up record per weld	H	H	S				
6	Preheat and interpass temperatures maintained during welding	WS-01, SW-WPS-2	Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F	Weld record per weld (welder, temperatures)	S	S	S				
7	Visual examination of every new weld	SW-NDE-1, SW-NDE-4	No crack, lack of fusion or incomplete penetration	VT report	W	W	W				
8	Volumetric examination per the NDE table; repairs re-examined	SW-NDE-2, SW-NDE-4	Acceptance per SW-NDE-4 and the owner's criteria	RT/UT reports	R	H	R				
9	Flanged ends: faces inspected, new gaskets, studs to preload	SW-NDE-5, SW-WO-5	No radial scratch; preload as stated	Bolt-up record	W	W	—				
10	Hydrostatic test at 210 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record with gauge calibration	H	H	H				
11	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
12	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Isometric ISO-6"-P150-P-1001

ISOMETRIC — 6"-P150-P-1001 — Replace spool SP02 of 6"-P150-P-1001



BILL OF MATERIALS

1	GASKET 6" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P150	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P150	4
4	FLANGE WELD-NECK 6" P150 RAISED FACE	3
5	ELBOW 90° R = 5/3 OD 6" P150	8
6	PIPE 6" P150 BEVELLED ENDS	174'-6-1/8"
7	TEE REDUCING 6" × 3/4" P150	2
8	TEE EQUAL 6" P150	1
9	REDUCER CONCENTRIC 6" × 4" P150	1
10	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1
11	FLANGE WELD-NECK 4" P150 RAISED FACE	1
12	GASKET 4" P150 COMPRESSED SHEET	1

WELDS 28 (4 FIELD) · SUPPORTS 9 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8F W9 W10
W11 W12 W13 W14 W15 W16F W17 W18 W19 W20
W21 W22 W23 W24 W25 W26F W27 W28

NEW WELDS (WO-100-03)

WO-100-03-W01 replaces W1
WO-100-03-W02 replaces W2
WO-100-03-W03 replaces W3
WO-100-03-W04 replaces W4
WO-100-03-W05 replaces W5
WO-100-03-W06 replaces W6
WO-100-03-W07 replaces W7
WO-100-03-W08 replaces W8

6"-P150-P-1001

CLASS H150 — Hydrocarbon process
SERVICE P · 6" · P150 · DESIGN 150 psi
T-101-N1 → P-101A-S
PIPE 174'-6-1/8" · WEIGHT 3040 lb

DWG ISO-6"-P150-P-1001 REV 0 WO-100-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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